

Stochastic Financial Models

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1 Martingales and Conditional Expectation

Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. All random variables are real-valued unless otherwise stated.

1.1 Probability spaces and sigma-algebras

Definition: sigma-algebra. A collection $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is a σ -algebra on Ω if:

1. $\Omega \in \mathcal{F}$;
2. if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
3. if $A_1, A_2, \dots \in \mathcal{F}$, then

$$\bigcup_{k=1}^{\infty} A_k \in \mathcal{F}.$$

These properties also imply that

$$\emptyset \in \mathcal{F}, \quad \bigcap_{k=1}^{\infty} A_k \in \mathcal{F} \quad \text{whenever } A_k \in \mathcal{F}.$$

Here $\mathcal{P}(\Omega)$ denotes the power set of Ω . Thus, $\mathcal{F}$ is a collection of subsets of Ω , and each $A \in \mathcal{F}$ is called an *event*.

A probability space consists of:

- a sample space Ω ;
- a σ -algebra $\mathcal{F}$ of events;
- a countably additive probability measure $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ with $\mathbb{P}(\Omega) = 1$.

Generated sigma-algebra. For a collection $\mathcal{C} \subseteq \mathcal{P}(\Omega)$, the notation $\sigma(\mathcal{C})$ denotes the smallest σ -algebra containing $\mathcal{C}$. It is the intersection of all σ -algebras on Ω that contain $\mathcal{C}$.

Almost sure equality. Two random variables U and V are equal *almost surely*, written $U = V$ a.s., if

$$\mathbb{P}(\{\omega \in \Omega : U(\omega) = V(\omega)\}) = 1.$$

Thus, they may differ on a set of probability zero.

1.2 Borel sets and measurability

Definition: Borel sigma-algebra. The *Borel σ -algebra* on $\mathbb{R}^d$ is

$$\mathcal{B}(\mathbb{R}^d) = \sigma(\{U \subseteq \mathbb{R}^d : U \text{ is open}\}).$$

It is the smallest σ -algebra containing every open subset of $\mathbb{R}^d$.

An element $B \in \mathcal{B}(\mathbb{R}^d)$ is called a *Borel set*. Borel sets include open sets, closed sets, singletons, countable sets, and countable unions or intersections of Borel sets.

For example,

$$(a, b), \quad (-\infty, a], \quad \{a\}, \quad \mathbb{Q}$$

are Borel subsets of $\mathbb{R}$. Moreover,

$$\mathcal{B}(\mathbb{R}) = \sigma(\{(-\infty, a] : a \in \mathbb{R}\}).$$

The notation distinguishes a single set from a collection:

$$B \subseteq \mathbb{R}^d, \quad B \in \mathcal{B}(\mathbb{R}^d), \quad \mathcal{B}(\mathbb{R}^d) \subseteq \mathcal{P}(\mathbb{R}^d).$$

Definition: measurable random variable. Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ -algebra. A function $X : \Omega \rightarrow \mathbb{R}$ is *$\mathcal{G}$ -measurable* if

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{G} \quad \text{for every } B \in \mathcal{B}(\mathbb{R}).$$

The expression $X^{-1}(B)$ denotes a preimage; it does not require X to have an inverse function.

Equivalently, it is enough to check that

$$\{X \leq a\} \in \mathcal{G} \quad \text{for every } a \in \mathbb{R}.$$

A real-valued *random variable* on $(\Omega, \mathcal{F}, \mathbb{P})$ is an $\mathcal{F}$ -measurable function.

Meaning of X_n being $\mathcal{F}_n$ -measurable. The statement means that

$$\{X_n \in B\} \in \mathcal{F}_n \quad \text{for every } B \in \mathcal{B}(\mathbb{R}).$$

If $\mathcal{F}_n$ represents the information available at time n , then the value of X_n is determined by that information. In particular, every question of the form “Is X_n in the Borel set B ?” can be answered using the information in $\mathcal{F}_n$.

Example: a sigma-algebra containing partial information. Let

$$\Omega = \{1, 2, 3, 4\}, \quad \mathcal{F} = \mathcal{P}(\Omega),$$

and consider

$$\mathcal{G} = \{\emptyset, \Omega, \{1, 2\}, \{3, 4\}\}.$$

This information distinguishes the two groups $\{1, 2\}$ and $\{3, 4\}$, but does not distinguish outcomes within either group.

The random variable

$$U(\omega) = \begin{cases} 0, & \omega \in \{1, 2\}, \\ 1, & \omega \in \{3, 4\} \end{cases}$$

is $\mathcal{G}$ -measurable.

However,

$$V(\omega) = \begin{cases} 0, & \omega \in \{1, 3\}, \\ 1, & \omega \in \{2, 4\} \end{cases}$$

is not $\mathcal{G}$ -measurable, because

$$\{V = 1\} = \{2, 4\} \notin \mathcal{G}.$$

In this finite example, a function is $\mathcal{G}$ -measurable precisely when it is constant on each of the two groups.

1.3 Indicators and integrability

Definition: indicator random variable. For an event $A \in \mathcal{F}$, define

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

Thus,

$$X\mathbf{1}_A = \begin{cases} X, & \text{on } A, \\ 0, & \text{on } A^c. \end{cases}$$

In particular,

$$\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A), \quad \mathbb{E}[X\mathbf{1}_A] = \int_A X \, d\mathbb{P}.$$

Definition: integrability. A random variable X is *integrable* if

$$\mathbb{E}[|X|] < \infty.$$

If X is integrable, then $X\mathbf{1}_A$ is integrable for every event A , since

$$|X\mathbf{1}_A| \leq |X| \implies \mathbb{E}[|X\mathbf{1}_A|] \leq \mathbb{E}[|X|] < \infty.$$

1.4 Filtrations and adapted processes

Definition: filtration. A *filtration* is a sequence $(\mathcal{F}_n)_{n \geq 0}$ of sub- σ -algebras of $\mathcal{F}$ such that

$$\mathcal{F}_n \subseteq \mathcal{F}_{n+1} \subseteq \mathcal{F}, \quad n \geq 0.$$

The σ -algebra $\mathcal{F}_n$ represents the information available at time n . The inclusions express that information accumulates over time.

Definition: stochastic process. A discrete-time *stochastic process* is a sequence

$$X = (X_n)_{n \geq 0}$$

of random variables $X_n : \Omega \rightarrow \mathbb{R}$.

Definition: adaptedness. The process X is *adapted* to $(\mathcal{F}_n)_{n \geq 0}$ if X_n is $\mathcal{F}_n$ -measurable for every $n \geq 0$.

The process is *integrable* if

$$\mathbb{E}[|X_n|] < \infty \quad \text{for every } n \geq 0.$$

This is a condition at each fixed time; it does not require $\sup_n \mathbb{E}[|X_n|] < \infty$.

1.5 Natural filtrations

Definition: natural filtration. For a stochastic process $X = (X_n)_{n \geq 0}$, its *natural filtration* is

$$\mathcal{F}_n^X = \sigma(X_0, X_1, \dots, X_n), \quad n \geq 0.$$

Here $\sigma(X_0, \dots, X_n)$ denotes the smallest σ -algebra making all of $X_0, \dots, X_n$ measurable.

More explicitly, define the random vector

$$W_n : \Omega \rightarrow \mathbb{R}^{n+1}, \quad W_n(\omega) = (X_0(\omega), \dots, X_n(\omega)).$$

Then

$$\mathcal{F}_n^X = \{W_n^{-1}(B) : B \in \mathcal{B}(\mathbb{R}^{n+1})\},$$

or, equivalently,

$$\mathcal{F}_n^X = \{\{\omega \in \Omega : (X_0(\omega), \dots, X_n(\omega)) \in B\} : B \in \mathcal{B}(\mathbb{R}^{n+1})\}.$$

Interpreting the notation. For each Borel set $B \subseteq \mathbb{R}^{n+1}$,

$$\{(X_0, \dots, X_n) \in B\}$$

is an event in Ω . It consists of the outcomes for which the observed vector belongs to B .

For example, when $n = 1$, take

$$B = (-\infty, 0] \times (1, \infty).$$

Then

$$\{(X_0, X_1) \in B\} = \{X_0 \leq 0\} \cap \{X_1 > 1\}.$$

The set B describes a condition on possible observed values; its preimage is the corresponding event in the sample space. Allowing B to range over all Borel sets gives the entire σ -algebra $\sigma(X_0, X_1)$.

Smallest-filtration property. The natural filtration is the smallest filtration to which X is adapted.

Indeed, suppose X is adapted to another filtration $(\mathcal{F}_n)_{n \geq 0}$. For $k \leq n$, the random variable X_k is $\mathcal{F}_k$ -measurable and hence $\mathcal{F}_n$ -measurable, since $\mathcal{F}_k \subseteq \mathcal{F}_n$. Therefore,

$$\mathcal{F}_n^X = \sigma(X_0, \dots, X_n) \subseteq \mathcal{F}_n.$$

1.6 Conditional probabilities and expectations

1.6.1 Conditioning on an event

Let $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. The conditional probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For fixed B , the map

$$A \longmapsto \mathbb{P}(A \mid B)$$

is a probability measure on $(\Omega, \mathcal{F})$.

For an integrable random variable X , define

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[X \mathbf{1}_B]}{\mathbb{P}(B)}.$$

This is a number: the average of X conditional on the event B occurring.

1.6.2 Conditioning on a sigma-algebra

Theorem: existence and almost sure uniqueness. Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ -algebra, and let X be integrable. There exists an integrable random variable Y satisfying:

1. Y is $\mathcal{G}$ -measurable;
2. for every $A \in \mathcal{G}$,

$$\mathbb{E}[Y \mathbf{1}_A] = \mathbb{E}[X \mathbf{1}_A].$$

Moreover, if Y' is another integrable random variable satisfying these conditions for the same X , then

$$Y = Y' \quad \text{a.s.}$$

Definition: conditional expectation. Any such Y is called a *version of the conditional expectation of X given $\mathcal{G}$* . We write

$$Y = \mathbb{E}[X \mid \mathcal{G}] \quad \text{a.s.}$$

The two defining conditions have distinct roles:

- Measurability requires Y to use only the information contained in $\mathcal{G}$.
- The integral identities require Y to preserve the average of X over every event observable using that information:

$$\int_A Y \, d\mathbb{P} = \int_A X \, d\mathbb{P}, \quad A \in \mathcal{G}.$$

Taking $A = \Omega$ gives

$$\mathbb{E}[Y] = \mathbb{E}[X].$$

However, this single equality is insufficient to characterize conditional expectation: the identity must hold for every $A \in \mathcal{G}$.

Why the notation includes “almost surely”. The theorem identifies a unique random variable up to almost sure equality. Two admissible versions may differ on a set of probability zero while satisfying exactly the same integral identities.

Thus, writing

$$Y = \mathbb{E}[X \mid \mathcal{G}] \quad \text{a.s.}$$

names the random variable characterized by the theorem. It is not an additional assumption on Y .

1.6.3 Monotonicity and the proof of uniqueness

Suppose X and X' are integrable and

$$X \leq X' \quad \text{a.s.}$$

Let Y and Y' satisfy the defining conditions of conditional expectation for X and X' , respectively, with respect to the same σ -algebra $\mathcal{G}$.

Set

$$A = \{Y > Y'\}, \quad Z = (Y - Y')\mathbf{1}_A.$$

Since Y and Y' are $\mathcal{G}$ -measurable,

$$A \in \mathcal{G}.$$

Also,

$$0 \leq Z \leq |Y| + |Y'|,$$

so Z is nonnegative and integrable.

Using the defining integral identities,

$$\begin{aligned} \mathbb{E}[Z] &= \mathbb{E}[(Y - Y')\mathbf{1}_A] \\ &= \mathbb{E}[Y\mathbf{1}_A] - \mathbb{E}[Y'\mathbf{1}_A] \\ &= \mathbb{E}[X\mathbf{1}_A] - \mathbb{E}[X'\mathbf{1}_A] \\ &= \mathbb{E}[(X - X')\mathbf{1}_A] \leq 0. \end{aligned}$$

Since $Z \geq 0$, we also have $\mathbb{E}[Z] \geq 0$. Hence

$$\mathbb{E}[Z] = 0.$$

A nonnegative random variable with expectation zero is zero almost surely, so $Z = 0$ a.s. Consequently,

$$Y \leq Y' \quad \text{a.s.}$$

This proves *monotonicity*:

$$X \leq X' \quad \text{a.s.} \implies \mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[X' \mid \mathcal{G}] \quad \text{a.s.}$$

For uniqueness, take $X' = X$. The argument gives $Y \leq Y'$ a.s. Interchanging Y and Y' gives $Y' \leq Y$ a.s. Therefore,

$$Y = Y' \quad \text{a.s.}$$

1.6.4 A finite-partition example

Suppose $A_1, \dots, A_m$ form a measurable partition of Ω , with $\mathbb{P}(A_i) > 0$, and let

$$\mathcal{G} = \sigma(A_1, \dots, A_m).$$

Then

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{i=1}^m \frac{\mathbb{E}[X \mathbf{1}_{A_i}]}{\mathbb{P}(A_i)} \mathbf{1}_{A_i}.$$

Thus, on each part of the partition, conditional expectation replaces X by its average over that part.

For the earlier example

$$\Omega = \{1, 2, 3, 4\}, \quad \mathcal{G} = \{\emptyset, \Omega, \{1, 2\}, \{3, 4\}\},$$

suppose all four outcomes are equally likely and $X(\omega) = \omega$. Then

$$\mathbb{E}[X \mid \mathcal{G}](\omega) = \begin{cases} \frac{1+2}{2} = \frac{3}{2}, & \omega \in \{1, 2\}, \\ \frac{3+4}{2} = \frac{7}{2}, & \omega \in \{3, 4\}. \end{cases}$$

This is a random variable whose value depends on which group is observed.

1.6.5 Conditioning on a random variable

For a real-valued random variable W , define

$$\sigma(W) = \{W^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}.$$

Conditioning on W means conditioning on the information generated by W :

$$\mathbb{E}[X \mid W] := \mathbb{E}[X \mid \sigma(W)].$$

1.6.6 Conditional probability given a sigma-algebra

For an event $A \in \mathcal{F}$, define

$$\mathbb{P}(A \mid \mathcal{G}) := \mathbb{E}[\mathbf{1}_A \mid \mathcal{G}].$$

This is a $\mathcal{G}$ -measurable random variable, with

$$0 \leq \mathbb{P}(A \mid \mathcal{G}) \leq 1 \quad \text{a.s.}$$

Its defining integral identity is

$$\mathbb{E}[\mathbb{P}(A \mid \mathcal{G}) \mathbf{1}_B] = \mathbb{P}(A \cap B), \quad B \in \mathcal{G}.$$

1.7 Martingales, submartingales, and supermartingales

Fix a filtration $(\mathcal{F}_n)_{n \geq 0}$.

Definition: martingale. A process $X = (X_n)_{n \geq 0}$ is a *martingale* with respect to $(\mathcal{F}_n)_{n \geq 0}$ if:

1. X is adapted: X_n is $\mathcal{F}_n$ -measurable for every n ;
2. X is integrable: $\mathbb{E}[|X_n|] < \infty$ for every n ;
3. the *martingale property* holds:

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n \quad \text{a.s., for every } n \geq 0.$$

The martingale property says that, given the information available at time n , the conditional expected value at the next time is the current value.

Definition: submartingale. An adapted, integrable process X is a *submartingale* if

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n \quad \text{a.s., for every } n \geq 0.$$

Definition: supermartingale. An adapted, integrable process X is a *supermartingale* if

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n \quad \text{a.s., for every } n \geq 0.$$

Thus, the conditional expected increment is respectively zero, nonnegative, or nonpositive:

$$\mathbb{E}[X_{n+1} - X_n \mid \mathcal{F}_n] \begin{cases} = 0, & \text{for a martingale,} \\ \geq 0, & \text{for a submartingale,} \\ \leq 0, & \text{for a supermartingale.} \end{cases}$$

1.8 Equivalent integral characterizations

Suppose X is adapted and integrable. Then:

$$\begin{aligned} X \text{ is a martingale} &\iff \mathbb{E}[X_{n+1} \mathbf{1}_A] = \mathbb{E}[X_n \mathbf{1}_A], \\ X \text{ is a submartingale} &\iff \mathbb{E}[X_{n+1} \mathbf{1}_A] \geq \mathbb{E}[X_n \mathbf{1}_A], \\ X \text{ is a supermartingale} &\iff \mathbb{E}[X_{n+1} \mathbf{1}_A] \leq \mathbb{E}[X_n \mathbf{1}_A], \end{aligned}$$

where each condition on the right must hold for *every* $n \geq 0$ and *every* $A \in \mathcal{F}_n$.

Proof of the martingale characterization. Fix n and let

$$Y = \mathbb{E}[X_{n+1} \mid \mathcal{F}_n].$$

By definition,

$$\mathbb{E}[Y \mathbf{1}_A] = \mathbb{E}[X_{n+1} \mathbf{1}_A], \quad A \in \mathcal{F}_n.$$

If $Y = X_n$ a.s., the required integral equality follows.

Conversely, suppose

$$\mathbb{E}[X_{n+1} \mathbf{1}_A] = \mathbb{E}[X_n \mathbf{1}_A], \quad A \in \mathcal{F}_n.$$

Since X_n is integrable and $\mathcal{F}_n$ -measurable, it satisfies the defining conditions for the conditional expectation of X_{n+1} given $\mathcal{F}_n$. Almost sure uniqueness therefore gives

$$X_n = \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \quad \text{a.s.}$$

Why the inequality characterizations also hold. Again let $Y = \mathbb{E}[X_{n+1} \mid \mathcal{F}_n]$. If $Y \geq X_n$ a.s., then

$$\mathbb{E}[(Y - X_n)\mathbf{1}_A] \geq 0 \quad \text{for every } A \in \mathcal{F}_n.$$

Conversely, suppose these integral inequalities hold. The event

$$A = \{Y < X_n\}$$

belongs to $\mathcal{F}_n$. On this event, $(Y - X_n)\mathbf{1}_A \leq 0$, while the assumed inequality gives

$$\mathbb{E}[(Y - X_n)\mathbf{1}_A] \geq 0.$$

Hence this expectation is zero, so

$$(X_n - Y)\mathbf{1}_{\{Y < X_n\}} = 0 \quad \text{a.s.}$$

Therefore $Y \geq X_n$ a.s. The supermartingale case follows by reversing the inequalities.

Consequence for ordinary expectations. Taking $A = \Omega$ shows that

$$\mathbb{E}[X_{n+1}] \begin{cases} = \mathbb{E}[X_n], & \text{for a martingale,} \\ \geq \mathbb{E}[X_n], & \text{for a submartingale,} \\ \leq \mathbb{E}[X_n], & \text{for a supermartingale.} \end{cases}$$

These conditions on ordinary expectations alone are not sufficient: the integral conditions must hold on every event in $\mathcal{F}_n$.

1.9 A martingale under its natural filtration

Proposition. If X is a martingale with respect to $(\mathcal{F}_n)_{n \geq 0}$, then it is also a martingale with respect to its natural filtration $(\mathcal{F}_n^X)_{n \geq 0}$.

Proof. The process X is adapted to its natural filtration, and each X_n remains integrable. Moreover,

$$\mathcal{F}_n^X \subseteq \mathcal{F}_n.$$

Thus, for every $A \in \mathcal{F}_n^X$, the original martingale property gives

$$\mathbb{E}[X_{n+1}\mathbf{1}_A] = \mathbb{E}[X_n\mathbf{1}_A].$$

The integral characterization proves that X is a martingale with respect to $(\mathcal{F}_n^X)_{n \geq 0}$.

1.10 Examples from independent random variables

Let $X_1, X_2, \dots$ be independent integrable random variables. Set

$$\mathcal{F}_0 = \{\emptyset, \Omega\}, \quad \mathcal{F}_n = \sigma(X_1, \dots, X_n), \quad n \geq 1.$$

Then X_{n+1} is independent of $\mathcal{F}_n$.

Define

$$S_0 = 0, \quad S_n = \sum_{i=1}^n X_i,$$

and

$$Z_0 = 1, \quad Z_n = \prod_{i=1}^n X_i.$$

1.10.1 Sums of independent mean-zero random variables

Suppose

$$\mathbb{E}[X_n] = 0 \quad \text{for every } n \geq 1.$$

Then $(S_n)_{n \geq 0}$ is a martingale with respect to $(\mathcal{F}_n)_{n \geq 0}$.

Proof. Each S_n is a measurable function of $X_1, \dots, X_n$, so S_n is $\mathcal{F}_n$ -measurable. Also,

$$\mathbb{E}[|S_n|] \leq \sum_{i=1}^n \mathbb{E}[|X_i|] < \infty.$$

For $A \in \mathcal{F}_n$, independence gives

$$\mathbb{E}[X_{n+1} \mathbf{1}_A] = \mathbb{E}[X_{n+1}] \mathbb{P}(A) = 0.$$

Therefore,

$$\begin{aligned} \mathbb{E}[S_{n+1} \mathbf{1}_A] &= \mathbb{E}[(S_n + X_{n+1}) \mathbf{1}_A] \\ &= \mathbb{E}[S_n \mathbf{1}_A] + \mathbb{E}[X_{n+1} \mathbf{1}_A] \\ &= \mathbb{E}[S_n \mathbf{1}_A]. \end{aligned}$$

The integral characterization shows that S is a martingale.

1.10.2 Products of independent nonnegative mean-one variables

Suppose

$$X_n \geq 0 \quad \text{a.s.}, \quad \mathbb{E}[X_n] = 1 \quad \text{for every } n \geq 1.$$

Then $(Z_n)_{n \geq 0}$ is a martingale with respect to $(\mathcal{F}_n)_{n \geq 0}$.

Proof. Each Z_n is a measurable function of $X_1, \dots, X_n$, so Z_n is $\mathcal{F}_n$ -measurable.

Nonnegativity and independence give

$$\mathbb{E}[|Z_n|] = \mathbb{E}[Z_n] = \prod_{i=1}^n \mathbb{E}[X_i] = 1.$$

Thus, Z_n is integrable.

For $A \in \mathcal{F}_n$, the random variable $Z_n \mathbf{1}_A$ is $\mathcal{F}_n$ -measurable and integrable. It is therefore independent of X_{n+1} . Consequently,

$$\begin{aligned} \mathbb{E}[Z_{n+1} \mathbf{1}_A] &= \mathbb{E}[X_{n+1} Z_n \mathbf{1}_A] \\ &= \mathbb{E}[X_{n+1}] \mathbb{E}[Z_n \mathbf{1}_A] \\ &= \mathbb{E}[Z_n \mathbf{1}_A]. \end{aligned}$$

The integral characterization shows that Z is a martingale.